

Islam I. Abdulaal

Integrated Photonics Researcher

✉ contact@iabdul-aal.me  iabdul-aal.me

 [linkedin.com/in/iabdul-aal](https://www.linkedin.com/in/iabdul-aal)  [@iabdul-aal](https://github.com/iabdul-aal)  [0009-0004-9300-3936](https://orcid.org/0009-0004-9300-3936)

RESEARCH INTERESTS

Integrated nanophotonics, quantum photonic sources, nonlinear on-chip devices, physics-informed inverse design, and wave-equation-constrained neural networks for photonic device optimization.

EDUCATION

B.Sc. in Electronics and Communications Engineering <i>Alexandria University, Faculty of Engineering</i>	Aug 2021 – Aug 2026 <i>Onsite</i>
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RESEARCH EXPERIENCE

Research Intern <i>NanoPhoto Lab, IMRE, A*STAR</i>	Sep 2025 – Present <i>Remote</i>
Research Intern <i>Advanced Labs Program, Egypt Scholars Inc.</i>	Jul 2025 – Sep 2025 <i>Remote</i>
Research Intern <i>OPST Group, SmartCI Research Center, Alexandria University</i>	Jul 2023 – Aug 2025 <i>Hybrid</i>

SELECTED PUBLICATIONS

*Citation metrics based on Google Scholar (*h-index*: 2, *citations*: 11), July 2026.

Journal Articles

[1] **Abdulaal, I. I.**, Elsayed, A. W. A., Abdelraouf, O. A. M., “Recent advances in terahertz-based bound states in the continuum metasurfaces for 6g communications and medtech,” *Journal of Optics*, 2026. DOI: [10.1088/2040-8986/ae8605](https://doi.org/10.1088/2040-8986/ae8605) [Online]. Available: <https://doi.org/10.1088/2040-8986/ae8605>

Preprints

[2] Labeab, A. H. M., Sallam, B., Elsayed, A. W., **Abdulaal, I. I.**, Kamal, A. M., Abdelraouf, O. A. M., “Beyond data-driven: How physics-informed neural networks are reshaping multi-physics design and discovery,” arXiv:2606.21945, 2026. [Online]. Available: <https://arxiv.org/abs/2606.21945>

HONORS, AWARDS, AND RESEARCH FUNDING

Dean’s Honors for Academic Excellence <i>Issuer: Alexandria University, Faculty of Engineering</i>	2021 – 2025
ICMTC Artificial Intelligence Contest (AIC-2) – 4th Place <i>Issuer: Military Technical College</i>	2024
Huawei ICT Skills Competition – National Finalist (AI Track) <i>Issuer: Huawei Technologies – MENA Region</i>	2023
NASA Space Apps Challenge – Global Nominee <i>Issuer: NASA</i>	2021

LEADERSHIP EXPERIENCE AND ACADEMIC SERVICE

Board Member <i>Optica Egypt Local Section</i>	Jul 2026 – Present <i>Hybrid</i>
Chairman <i>IEEE Solid-State Circuits Society (SSCS), Alexandria University Student Branch Chapter</i>	Feb 2025 – Oct 2025 <i>Hybrid</i>
General Coordinator <i>Education Clinic (USA-Registered NGO)</i>	Aug 2021 – Sep 2023 <i>Remote</i>

TEACHING AND MENTORING EXPERIENCE

Tutor / Mentor, Optica Egypt	Jul 2026 – Present
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PROFESSIONAL MEMBERSHIPS

- **Optica (formerly OSA)** Jul 2026 – Present
Student Member.
- **Institute of Electrical and Electronics Engineers (IEEE)** 2025 – 2026
Student Member (ID: 101099759).

REFERENCES

Dr. Omar Abdelraouf

Research Scientist

Institute of Materials Research and Engineering (IMRE),
A*STAR, Singapore

✉ omar_abdelrahman@a-star.edu.sg

Prof. Ishac Kandas

Associate Professor

Department of Engineering Physics and Mathematics,
Alexandria University

✉ ishac.kandas@alexu.edu.eg